

Evaluation Report & Next-Step Execution Roadmap

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Assigned Construct: Crystallized Intelligence (Gc)

Proposed Task: Knowledge Application Task

Evaluation Status: Proceed with Modifications

1. Executive Summary

The proposed Knowledge Application Task is engaging and encourages students to connect knowledge with real-world situations. However, in its current form, the task moves beyond Crystallized Intelligence (Gc) and begins to assess Fluid Intelligence (Gf), critical thinking, problem solving, and communication skills. To align with our startup objectives, the task must primarily measure acquired knowledge, verbal comprehension, conceptual understanding, and vocabulary while minimizing construct contamination.

2. Alignment with MentiScope's Objectives

- Alignment with CHC-Gc Theory: 7/10
- Scientific Validity: 7/10
- Innovation: 8/10
- Adaptive Potential: 8/10
- AI Readiness: 8/10
- MVP Feasibility: 8/10
- Overall Evaluation: 7.5/10 – Proceed with Modifications

3. Scientific Assessment

Primary Gc components to measure: Vocabulary, verbal comprehension, general knowledge, conceptual understanding, and verbal reasoning.

Potential construct contamination: Fluid Intelligence (Gf), problem-solving ability, creativity, and communication skills.

4. Recommended Task Architecture

Stage 1 – Knowledge Activation: Present real-world scenarios followed by multiple-choice questions that activate prior knowledge.

Stage 2 – Concept Mapping: Students connect Cause → Effect → Solution relationships using guided mappings.

Stage 3 – Short Justification: Students provide a brief explanation (maximum 50 words) to justify their understanding.

This structure preserves knowledge measurement while reducing dependence on creativity and open-ended problem solving.

5. AI Analytics to Capture

- Knowledge accuracy
- Vocabulary richness
- Conceptual connections
- Reasoning coherence
- Response time
- Error patterns

6. MVP Deliverables (Current Phase)

- ✓ Scenario Bank (minimum 100 scenarios)
- ✓ Automated Rubric Engine
- ✓ NLP Scoring Pipeline
- ✓ Event Logger
- ✓ Analytics Output
- ✓ Documentation
- ✓ Pilot Dataset

7. Assessment SDK Compliance Requirements

All outputs must follow the common Assessment SDK specification.

Session Data: student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data: event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output: raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output: behavioral_metrics and recommendations.

8. Next Sprint Execution Plan (2 Weeks)

Sprint Objective: Redesign the task architecture to improve construct purity and prepare an MVP prototype.

Expected Deliverables:

1. Literature review on Crystallized Intelligence (Gc)
2. Construct map and sub-construct definitions
3. Scenario generator prototype
4. Concept mapping interface prototype
5. Scoring rubric and NLP evaluation plan
6. Event logging implementation
7. Sample analytics dashboard

9. Definition of Done (DoD)

Scientific: Gc construct clearly defined, construct contamination minimized, and sub-scores identified.

Technical: Prototype functional, generates unique scenarios, captures standardized behavioral data, and exports results in SDK format.

Validation: Human-AI scoring agreement $\geq 85\%$, bias testing across academic streams, and pilot study with 50–100 students.

Guidance to the Intern

Your responsibility is not to build the entire platform at this stage. Your responsibility is to build a scientifically valid micro-assessment engine for Crystallized Intelligence that follows the common Assessment SDK. The module will later be integrated into the unified cognitive and career assessment platform.